EXAM SHIELD

Data Science and Databases Question Bank

SQL, database design, transactions, warehousing and analytics.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 22 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Data Science and Databases

SQL, database design, transactions, warehousing and analytics.

1. Which SQL command removes rows from a table?

A. ERASE
B. REMOVE
C. DROP
D. DELETE

Answer: D

2. Which SQL keyword removes duplicate rows from a query result?

A. UNIQUE ROWS
B. SINGLE
C. ONLY
D. DISTINCT

Answer: D

3. Which normal form removes transitive dependencies?

A. Second normal form
B. First normal form
C. Third normal form
D. Unnormalised form

Answer: C

4. Which join returns every row of the left table plus matching rows from the right table?

A. CROSS JOIN
B. INNER JOIN
C. SELF JOIN
D. LEFT JOIN

Answer: D

5. Which kind of database stores records as flexible JSON-like documents?

A. Spreadsheet
B. Document database
C. Relational database
D. Flat file

Answer: B

6. In data engineering, ETL stands for:

A. Evaluate, Test, Launch
B. Encrypt, Transfer, Log
C. Export, Train, Learn
D. Extract, Transform, Load

Answer: D

7. Which measure of central tendency is least affected by outliers?

A. Median
B. Range
C. Mean
D. Variance

Answer: A

8. An index on a column mainly speeds up:

A. Searching and filtering on that column
B. Taking backups
C. Changing the schema
D. Inserting new rows

Answer: A

9. In SQL, rows that share a value are combined with GROUP _____, and the resulting groups are filtered with _____. [2 marks]

Answer: BY | HAVING

10. A foreign key enforces _____ integrity between two tables. [1 mark]

Answer: referential

11. In a data-warehouse star schema, a central _____ table is surrounded by dimension tables. [1 mark]

Answer: fact

12. Explain the ACID properties of a transaction, with an example of what goes wrong without each. [5 marks]

13. Describe the stages of a data-science project from problem definition to deployment, and the role of data cleaning. [5 marks]